

AMIT CHAKRABORTY

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Dhaka, Bangladesh

EDUCATION

North South University

Bachelor of Science in Computer Science and Engineering
CGPA: 3.26/4.00

Sep 2021 – Dec 2025
Dhaka, Bangladesh

EXPERIENCE

Machine Intelligence Lab, North South University

Research Assistant (Large Language Models (LLMs) & Applied Research)

May 2025 – Present
Dhaka, Bangladesh

- Built **PANDA** in collaboration with [LifeSpring](#), with voice integration to enable conversational **ADHD** screening, providing real-time feedback, reducing diagnosis time from several months to 45 minutes, and covering 27 DSM-5-based questions per session.
- Developed **LLM Weight Refinement Pruning**, a post-training LLM compression framework supporting unstructured pruning, hardware-efficient N:M structured sparsity, activation-aware importance scoring, and low-rank SVD decomposition for efficient inference and evaluation.

SKILLS

Languages & Frameworks: Python, JavaScript, Java, FastAPI, Express.js

Machine Learning & AI: PyTorch, Scikit-learn, HuggingFace Transformers, LangChain, LLaIndex, Ollama, RAG, LLM/SLM Fine-tuning, Prompt Engineering

Data & Visualization: Pandas, NumPy, Matplotlib, Seaborn

Database & DevOps: MySQL, MongoDB, VectorDB (FAISS, ChromaDB)

Tools & Productivity: GitHub, Trello, JSDoc

PROJECTS

LLM Weight Refinement Pruning | Python, PyTorch, Transformers, LLM Compression, N:M Structured Sparsity, Low-Rank SVD

- Developed a post-training LLM compression framework supporting both unstructured pruning and hardware-efficient N:M structured sparsity techniques.
- Introduced an activation-aware ClearCut importance scoring mechanism with optional channel reallocation, alongside attention-layer low-rank decomposition using Singular Value Decomposition (SVD); evaluated model performance using perplexity metrics on WikiText2, C4, and PTB, and conducted zero-shot evaluations on BoolQ, RTE, HellaSwag, ARC-Challenge, and MNLI.
- Project GitHub: [LLM Weight Refinement Pruning GitHub](#)

LLM Physics chatbot(RAG-Based) | SLMs, Python, Ollama, LangChain, HuggingFace, FastAPI, Prompt Engineering, Git

- Built a lightweight Retrieval-Augmented Generation (RAG) chatbot specifically for Physics education, designed for offline and low-resource environments.
- Achieved **75% accuracy** on a 4,000-question benchmark using RAGChecker, enabling accessible and device-efficient learning support for underserved communities.
- Project GitHub: [RAG Chatbot GitHub](#)

PANDA: Personalized ADHD Neuro Diagnostic Assessment | Next.js, TypeScript, Convex, PostgreSQL, OpenAI (GPT-4.1), STT/TTS, LLM Evaluation, Conversational AI, Large Language Models

- PANDA is an AI-driven ADHD screening system designed to provide an early indication of ADHD likelihood through a structured, **DSM-5**-based conversational assessment.
- Developed in collaboration with Lifespring, PANDA reduces the traditional diagnostic screening time by approximately **70–80%**, delivering a structured preliminary evaluation within 40–45 minutes.
- The system integrates both chat and voice interactions to ensure accessibility, efficiency, and research-backed assessment support.
- Project GitHub: [PANDA GitHub](#)

Health AI Assistant: Personalized Nutrition & Grocery Planner | Next.js, Python, Flask, Groq (Llama 3 & Whisper), STT/TTS, Conversational AI, Large Language Models

- An AI-driven health and grocery assistant designed to provide personalized dietary recommendations, budget-friendly meal planning, and BMI-based nutritional insights through an interactive conversational interface.
- Delivers highly tailored and localized guidance by integrating affordable, culturally relevant food options (e.g., Bengali cuisine), structured meal templates, and accurate per-meal calorie tracking to support diverse health goals.
- The system seamlessly integrates both text chat and real-time voice interactions (utilizing Groq's Whisper model for Speech-to-Text and browser-based Text-to-Speech) to ensure accessibility and an intuitive user experience.
- Project Links: [GitHub](#) | [Live Demo](#)

NeuroNote: AI-Powered Study Material Generator | React, Vite, Node.js, Express, MongoDB, Google Gemini API, Groq, JWT

- An AI-driven educational web application designed to instantly transform diverse input sources—including YouTube URLs, web articles, PDF document uploads, and plain text—into highly organized study materials like interactive flashcards, quizzes, detailed notes, and summaries.
- Developed a robust Node.js and Express backend integrated with MongoDB to handle data persistence and JWT-based user authentication, supporting both registered users with private cloud libraries and guest users utilizing temporary local storage.
- Engineered an adaptable system architecture allowing dynamic integration of multiple Large Language Models (including Google Gemini

- and Groq APIs), providing users with flexible API key configuration and customizable settings to tailor study material generation.
- Project Links: [GitHub](#) | [Live Demo](#)

DOMAIN OF INTEREST

AI, LLMs, and Applied Machine Learning:

- Large Language Models (LLMs)
- Natural Language Processing (NLP)
- Retrieval-Augmented Generation (RAG) & Evaluation
- Efficient & Edge Deployment of LLMs
- AI Tooling, and Low-Resource Solutions

EXTRACURRICULAR

- Top 15 Finalist (out of 100+ teams), NSU Capstone Project Challenge– Innovation Challenge 2025**, North South University
- Bear Summit and National Semiconductor Symposium 2025**, represented MILab, North South University
- Ex Probationary Member, NSU ACM Student Chapter**

REFERENCES

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